

BIOMECHANICS (EPHE 341)

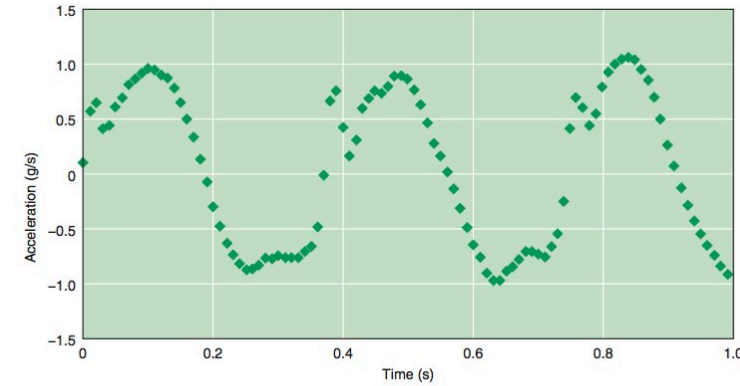
Signals

Separating the true signal from the noise.

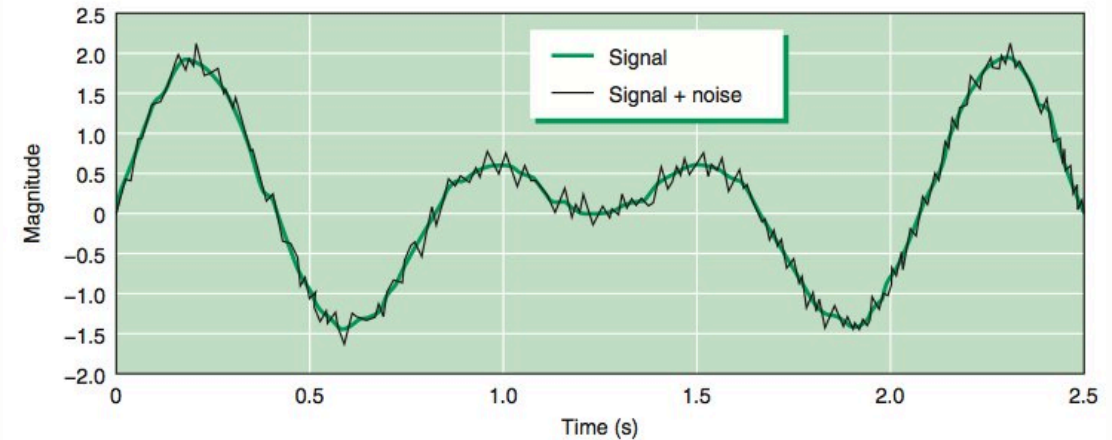
In biomechanics we are dealing with values from sensors that have limitations based on

- Sampling
- Sensor error
- Environmental error

We need to be able to separate the **true signal** from the **noise**



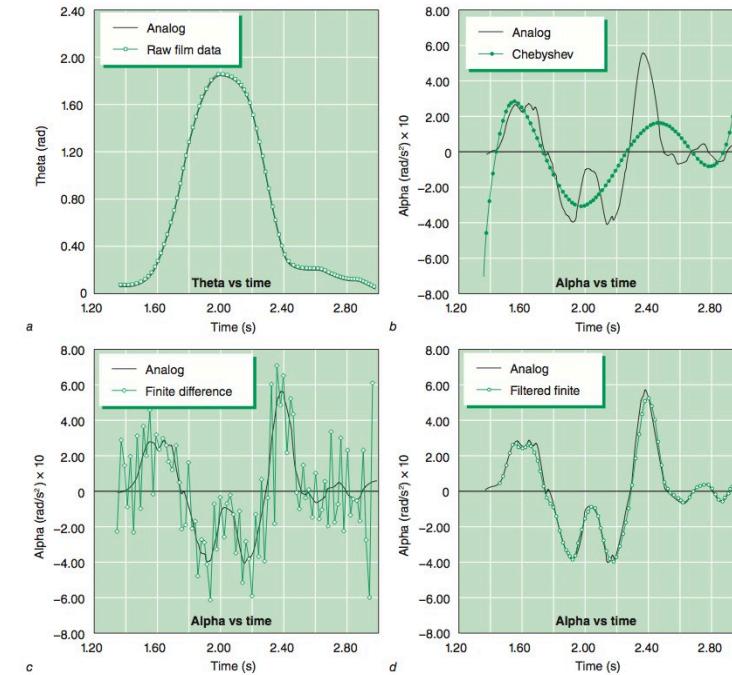
▲ **Figure 12.1** The digitized discrete representation of the acceleration of the head while a subject is running. The signal was sampled at 100 Hz (100 samples per second).



We can minimize the noise by **processing** the data.

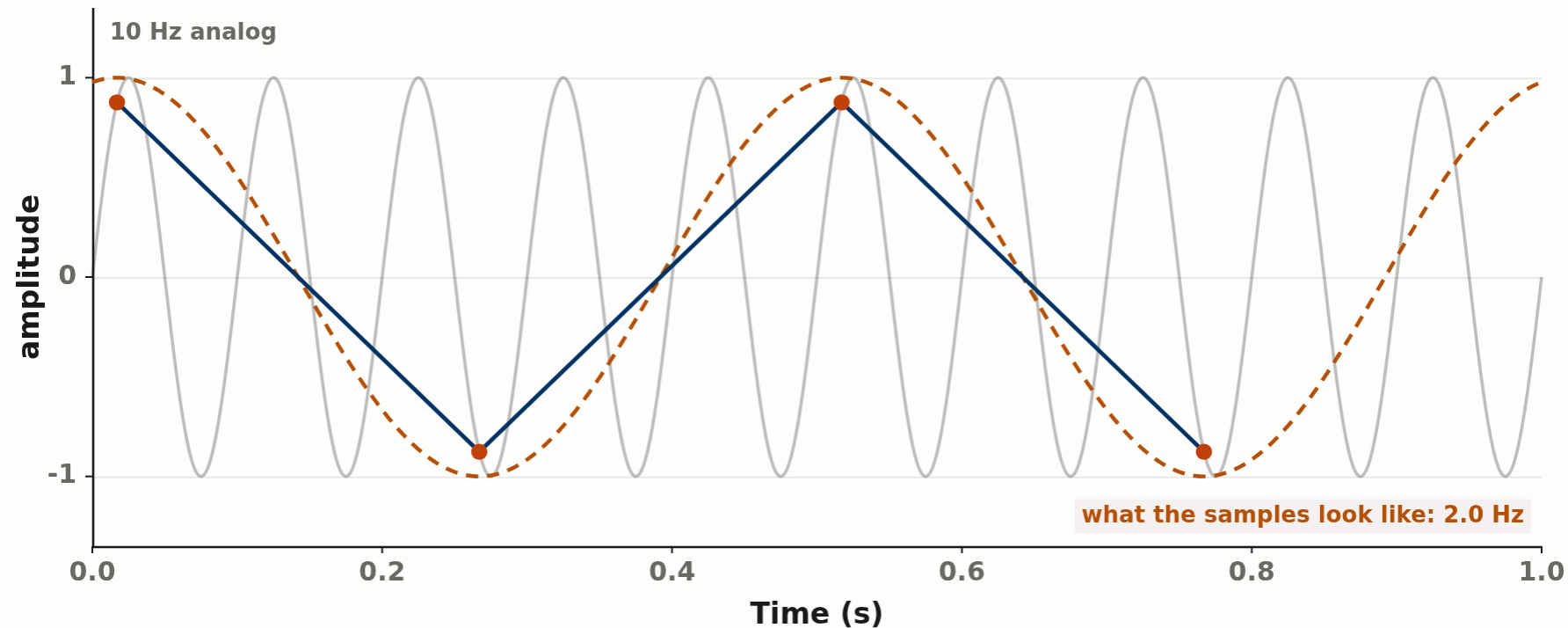
- Sampling properly
- Smoothing data
- Filtering data
- Fitting data

To inform our processing choices we need to understand some basic characteristics of signals and the transformations that we intend to perform on them.



Nyquist–Shannon Sampling Theorem

In sampling an analog signal, the goal is to represent the original signal as accurately as possible with as few discrete points as possible. **How high must the sampling frequency be for an analog signal to be accurately represented by discrete data points?**

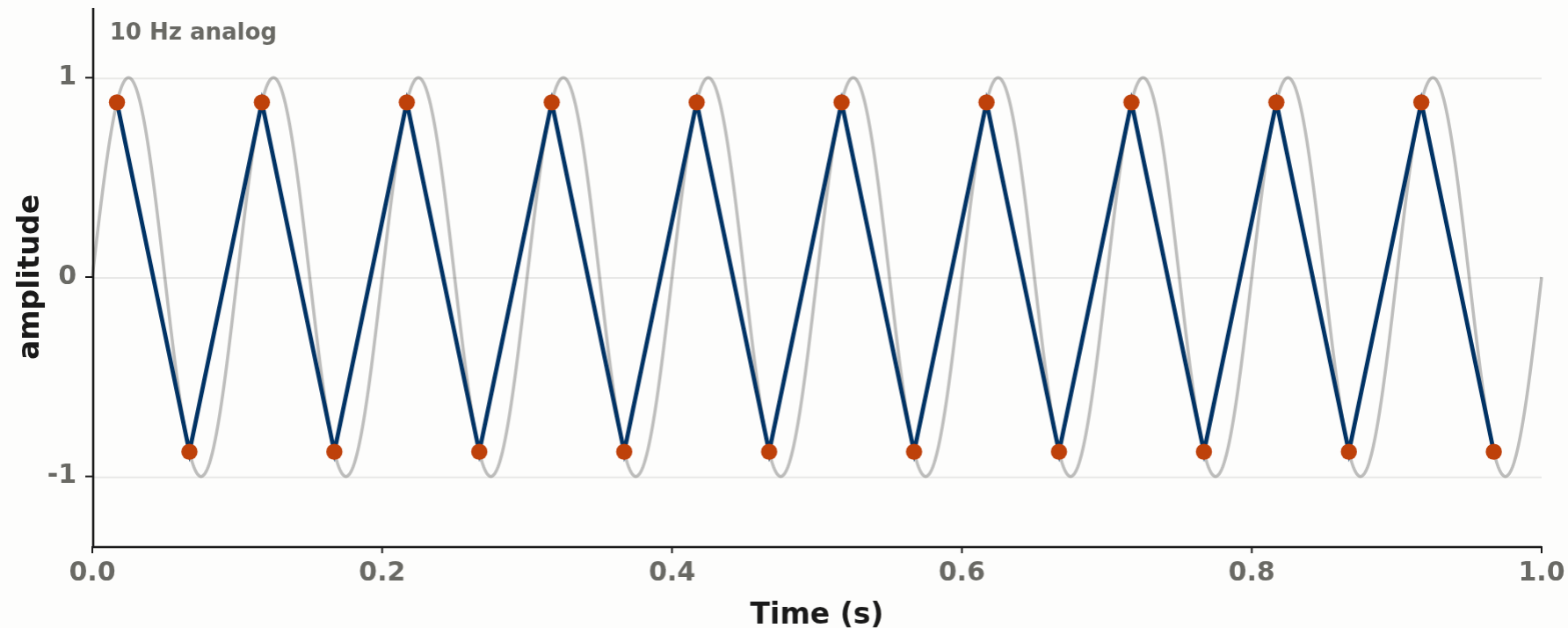


sampling at **4 Hz** · 4 samples in a second · $0.4\times$ the signal frequency

below the Nyquist frequency — **aliased**. The dashed line is a **different** sine wave that fits every sample exactly as well. Nothing in the data can tell them apart.

Nyquist–Shannon Sampling Theorem

For a digital signal to represent an analog signal perfectly, the sampling frequency must be **at least twice** the analog signal's highest frequency.



sampling at **20 Hz** · 20 samples in a second · 2.0× the signal frequency
above Nyquist — the shape is right but sharp turns are rounded off. The Nyquist frequency here is 20 Hz.

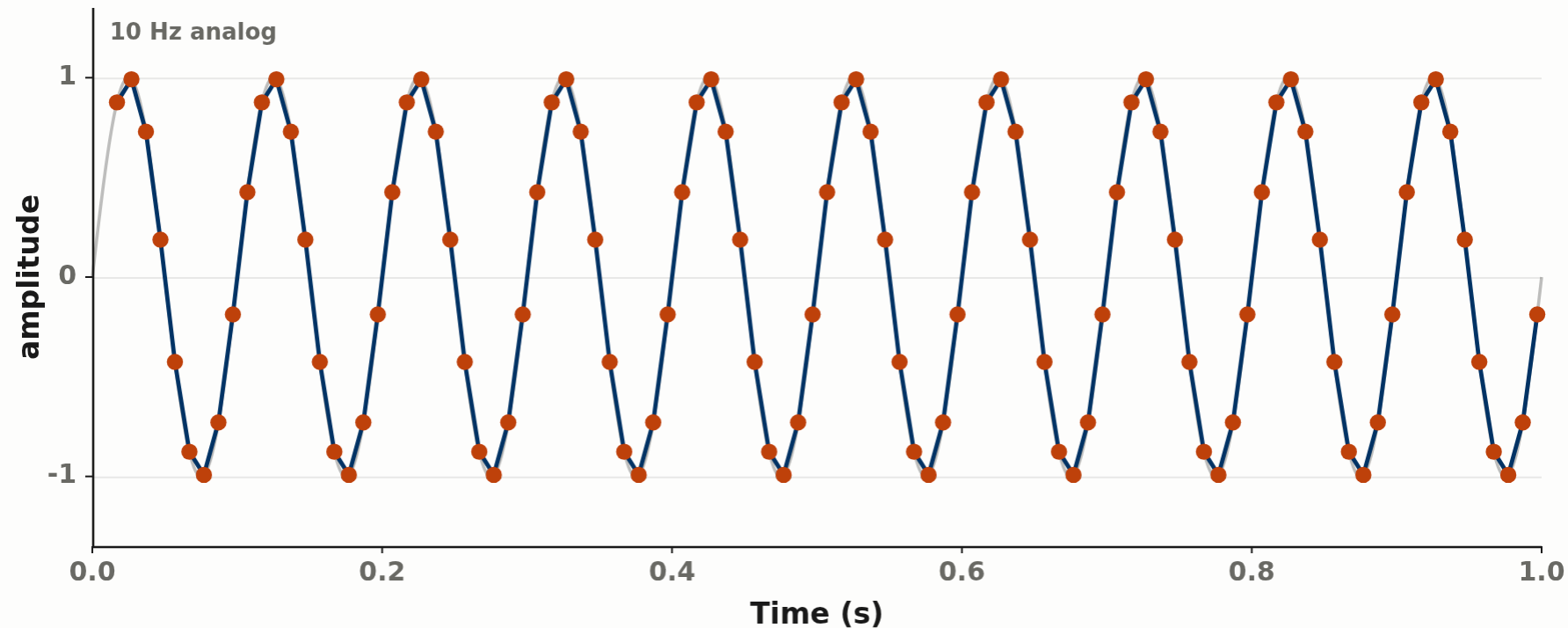
Sampling
frequency



20 Hz

Nyquist–Shannon Sampling Theorem

To ensure that sharp direction changes in the signal are not missed, sampling frequencies of **5–10 times** the Nyquist Frequency are often used.



sampling at **100 Hz** · 99 samples in a second · 10.0× the signal frequency
5–10× the signal frequency — direction changes are caught. The Nyquist frequency here is 20 Hz.

Sampling
frequency

100 Hz

Following data collection, “raw” data often contains noise.

Processing data post collection can improve the **signal to noise (S/N) ratio**.

Smoothing, filtering and curve fitting can improve S/N ratio.

Processing is necessary **before** calculations are performed.



Filtering, smoothing and fitting can improve S/N using different approaches.

Filtering:

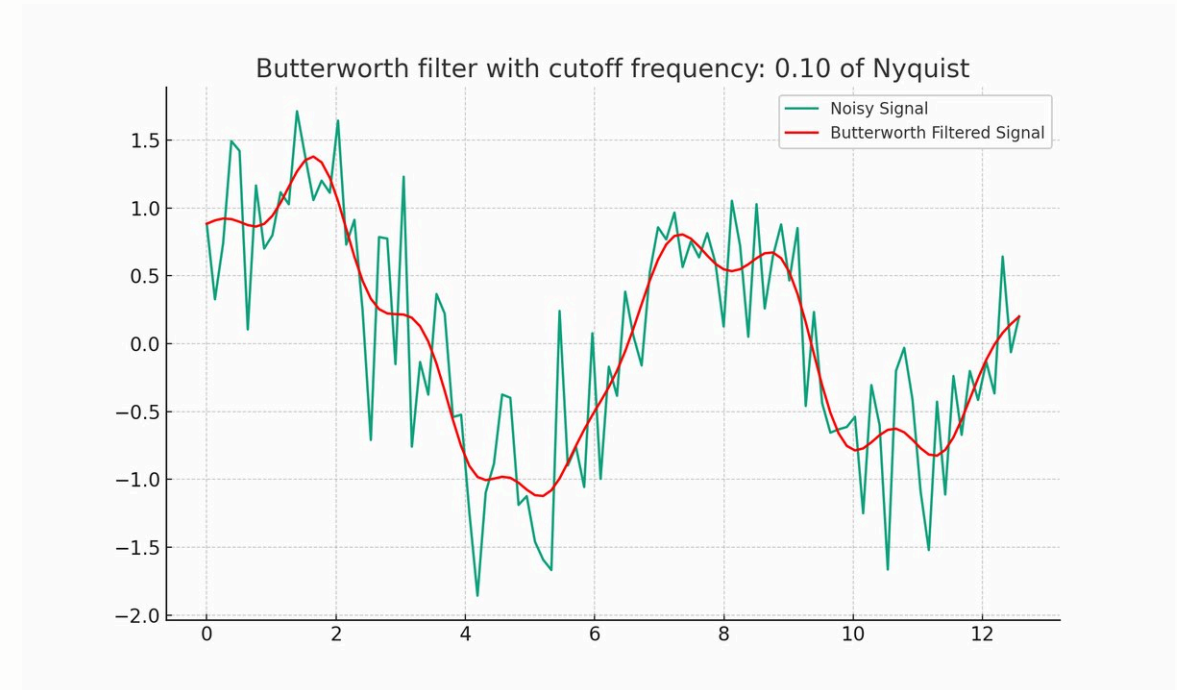
involves altering or removing certain frequency components from a signal.

Smoothing:

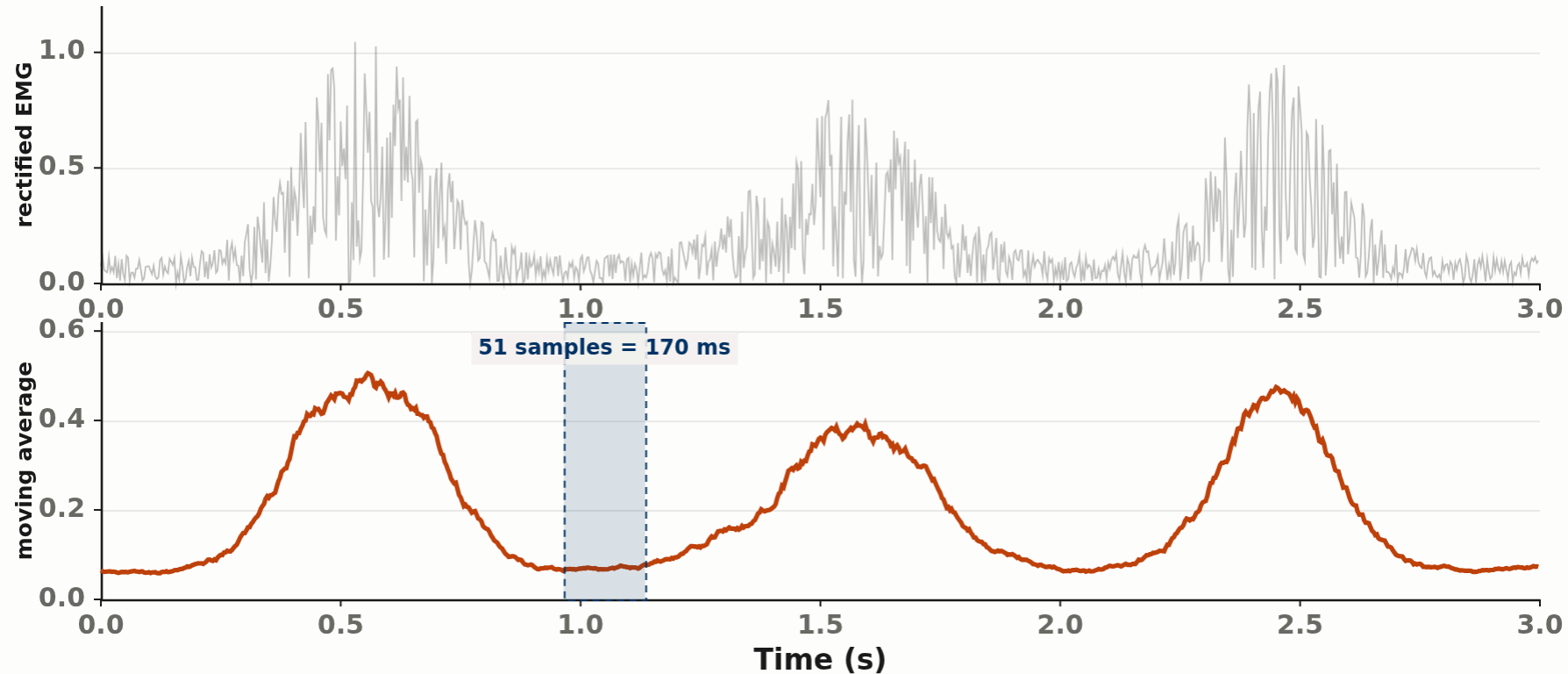
involves averaging or aggregating data points to reduce short-term fluctuations and highlight longer-term trends or cycles.

Curve Fitting:

to find a mathematical function that best represents a dataset. This function may be a simple linear equation, a complex polynomial, or any other



Moving average (over a window of samples). Simple averaging gets rid of noisy spikes.



window of **51** samples · 170 ms at this sample rate

A wider window is a smoother line — and a later, lower, blunter one. Watch the height of the bursts fall and their edges soften as the window grows past the length of the burst itself. Simple averaging gets rid of noisy spikes, but it cannot tell a spike from a fast real change.

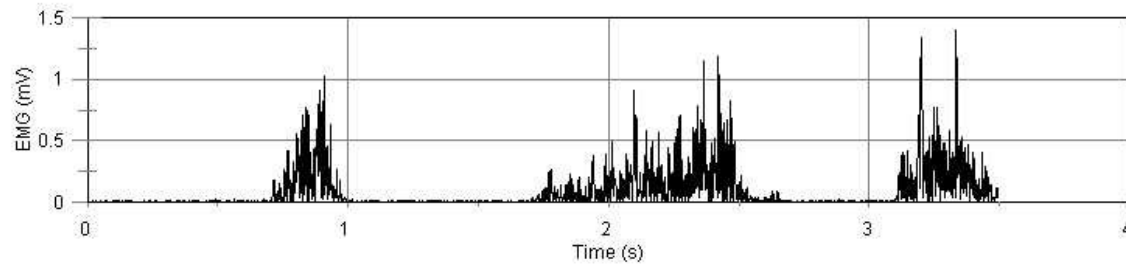
Window



51 pts

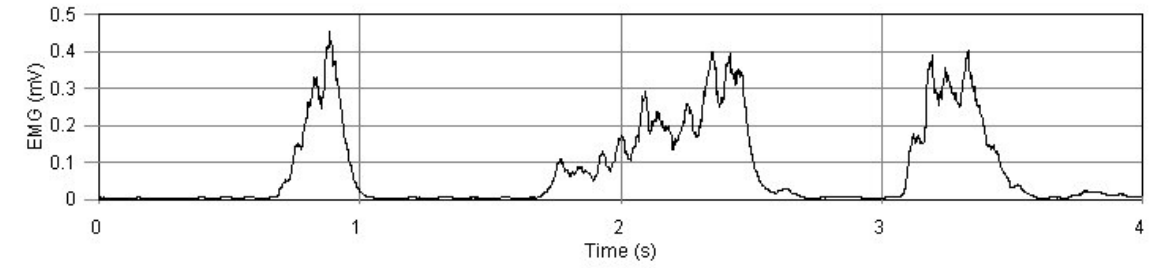
Smoothing

Full-wave rectified EMG



EMG (1010 Hz sampling rate)

Averaged EMG (window of 51)

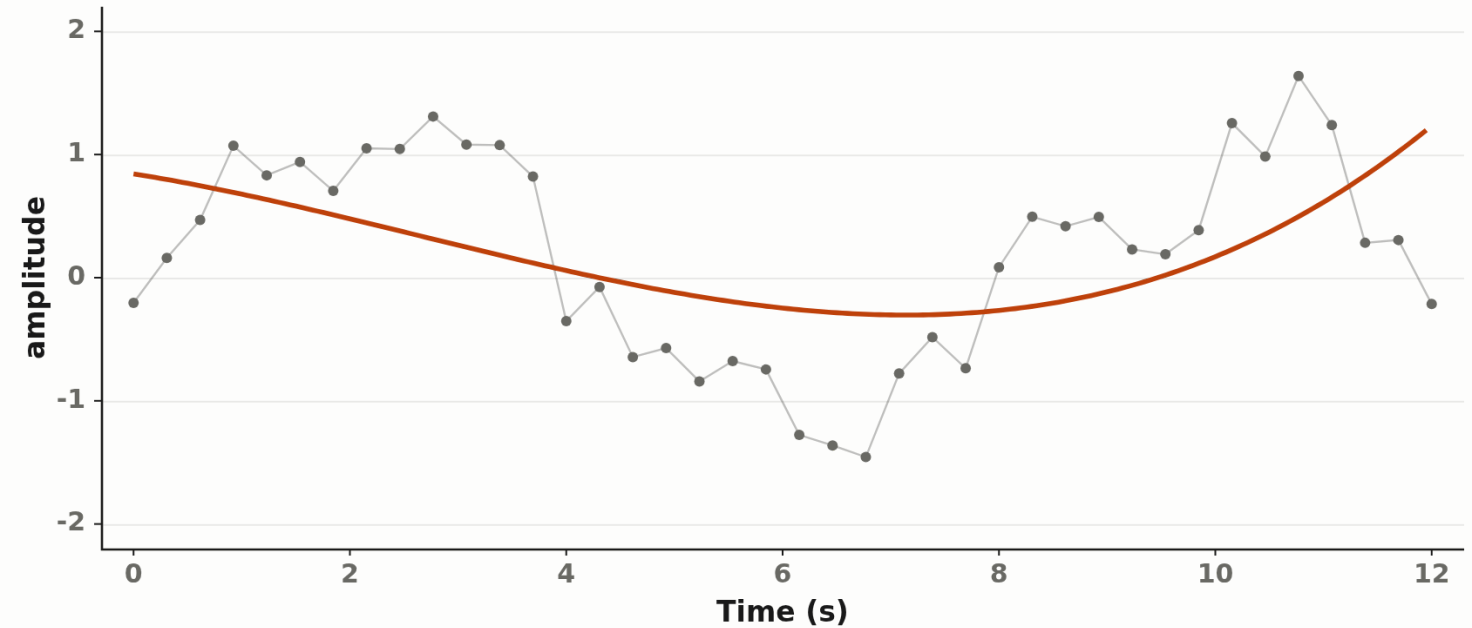


averaged EMG (moving average, 51 points)

Curve fitting

Fits a **best-fitting function** to the data.

- Polynomials
- Splines
- linear



polynomial of degree 3 · residual RMS **0.690**

Raise the order and the residual always falls — that is arithmetic, not insight. Somewhere past a handful of terms the curve stops describing the signal and starts describing this particular sample of noise.

linear

polynomial

spline

Polynomial order

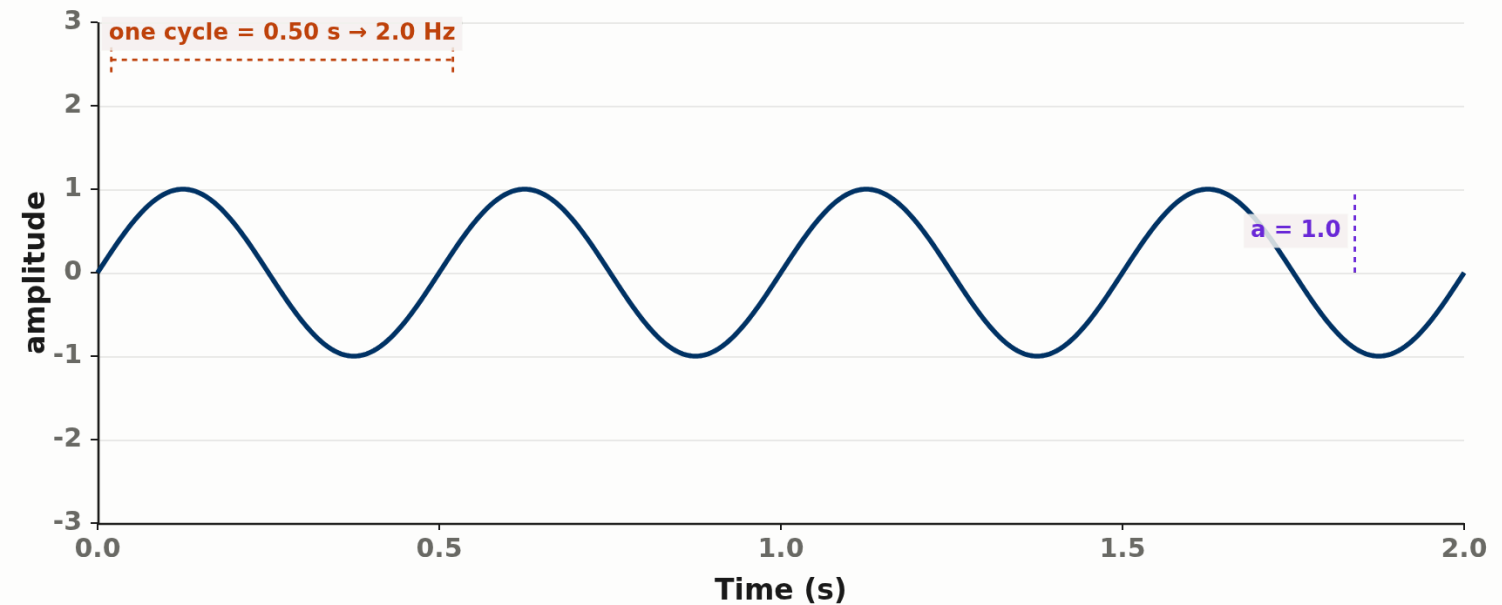


3

Components of a signal

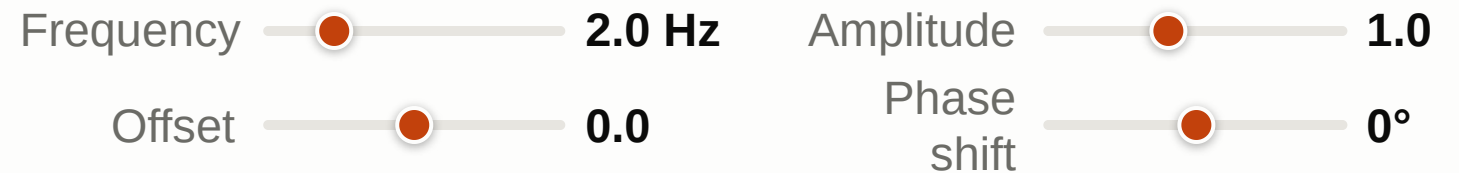
Filtering – a process that removes some unwanted components or features from a signal. Before we discuss filtering we need to talk about **signal components**.

- **frequency:** the number of repeating cycles per unit time
- **amplitude:** the magnitude of change
- **offset:** the distance of the mean amplitude from zero
- **Phase Shift:** a timing offset

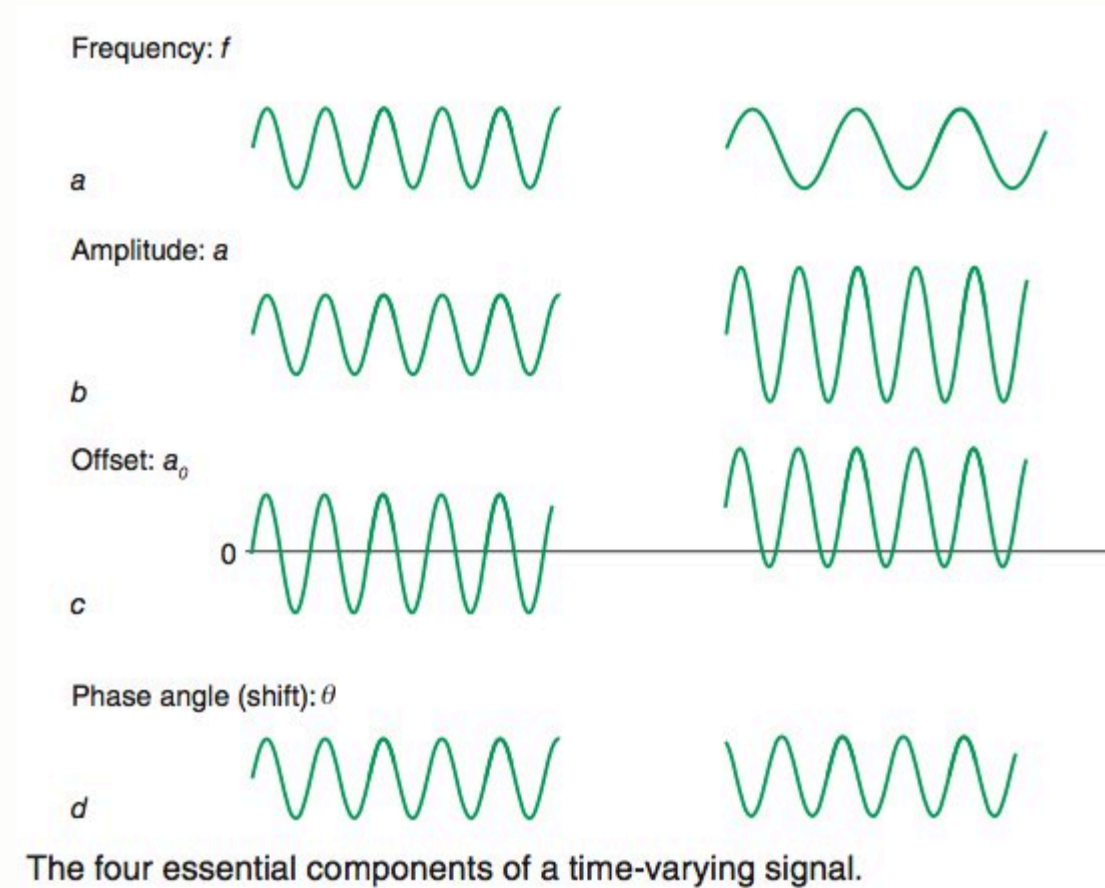


$$y(t) = 0.0 + 1.0 \cdot \sin(2\pi \cdot 2.0 \cdot t + 0^\circ)$$

The dashed grey wave is the starting signal. Frequency stretches it along time, amplitude stretches it up and down, offset lifts the whole thing off zero, and the phase shift slides it sideways without changing its shape.



Components of a signal



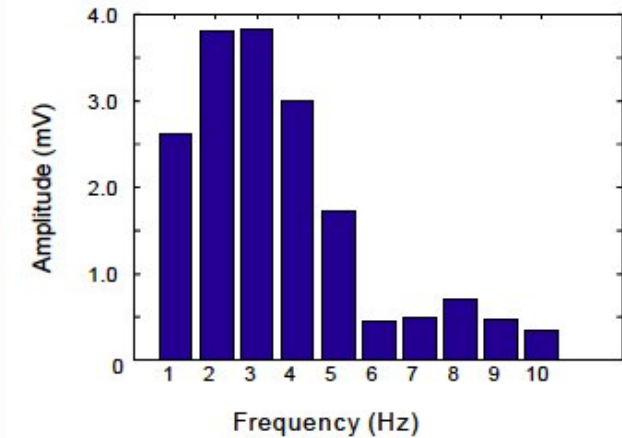
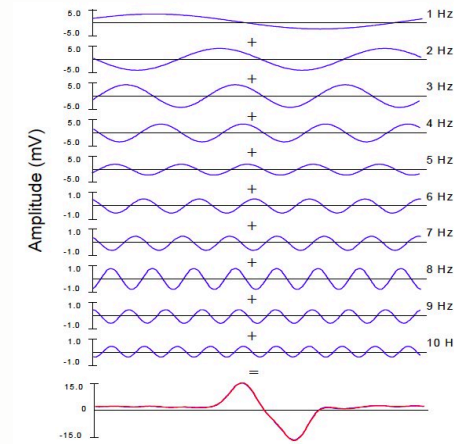
The four essential components of a time-varying signal.

Signals

Signals can be **decomposed and reconstructed** from a number of basic functions (sine waves).

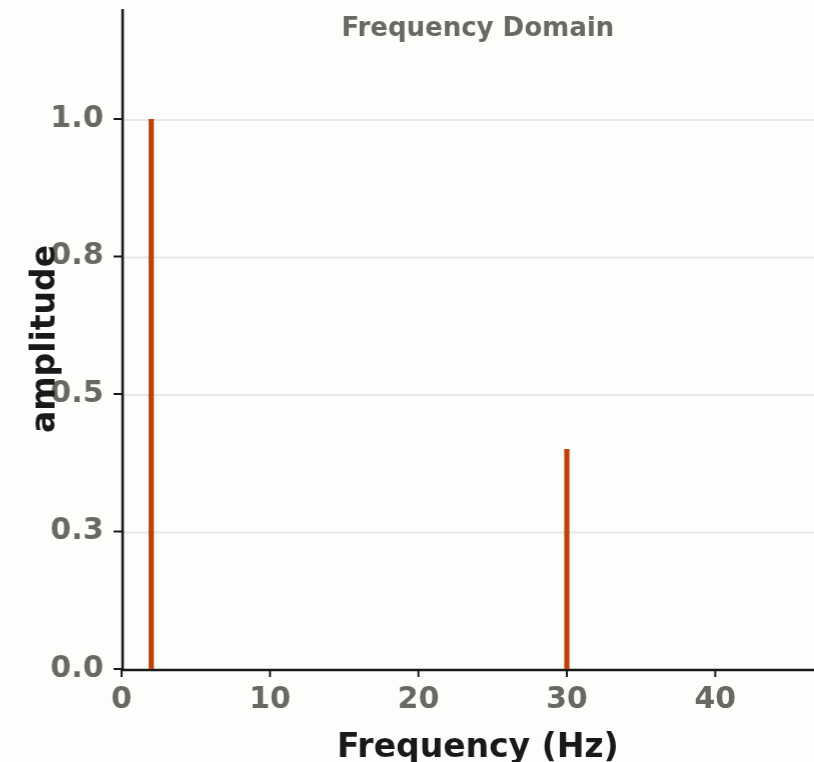
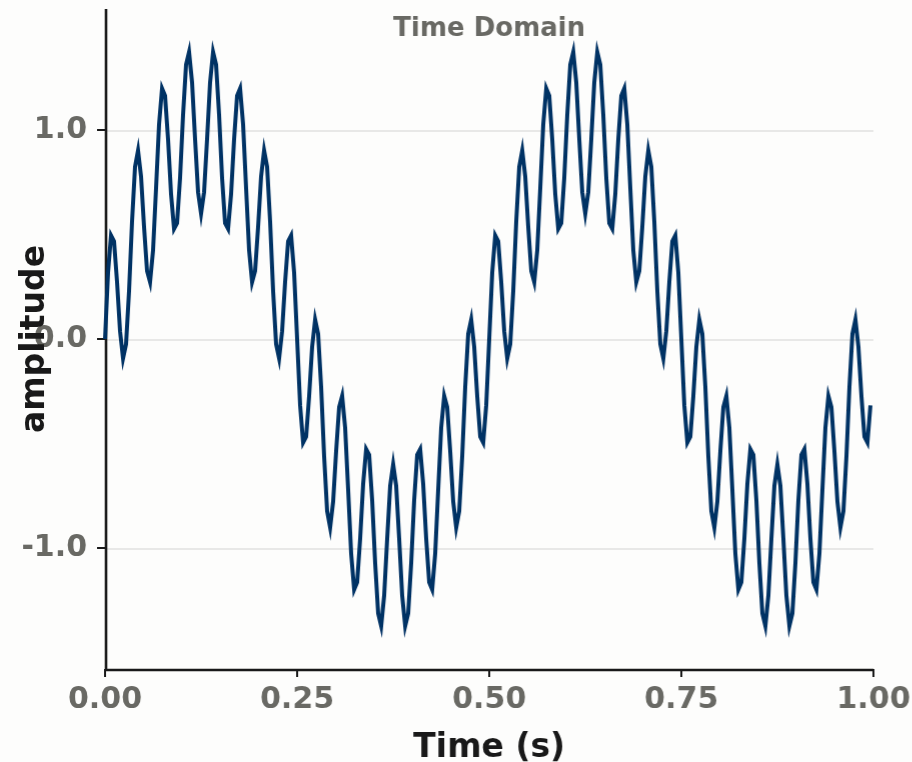
Once a signal is decomposed it can be represented in a frequency graph of the basic sine waves.

Time Domain → Frequency Domain



Frequency domain

A time-domain signal is converted to a frequency-domain signal with a mathematical technique called a **Fourier Transform**.



2 Hz + 30 Hz

The same signal, said twice. On the left, amplitude against time — what the sensor recorded. On the right, amplitude against frequency — which sine waves it is made of. A Fourier transform is the arithmetic that turns the first into the second.

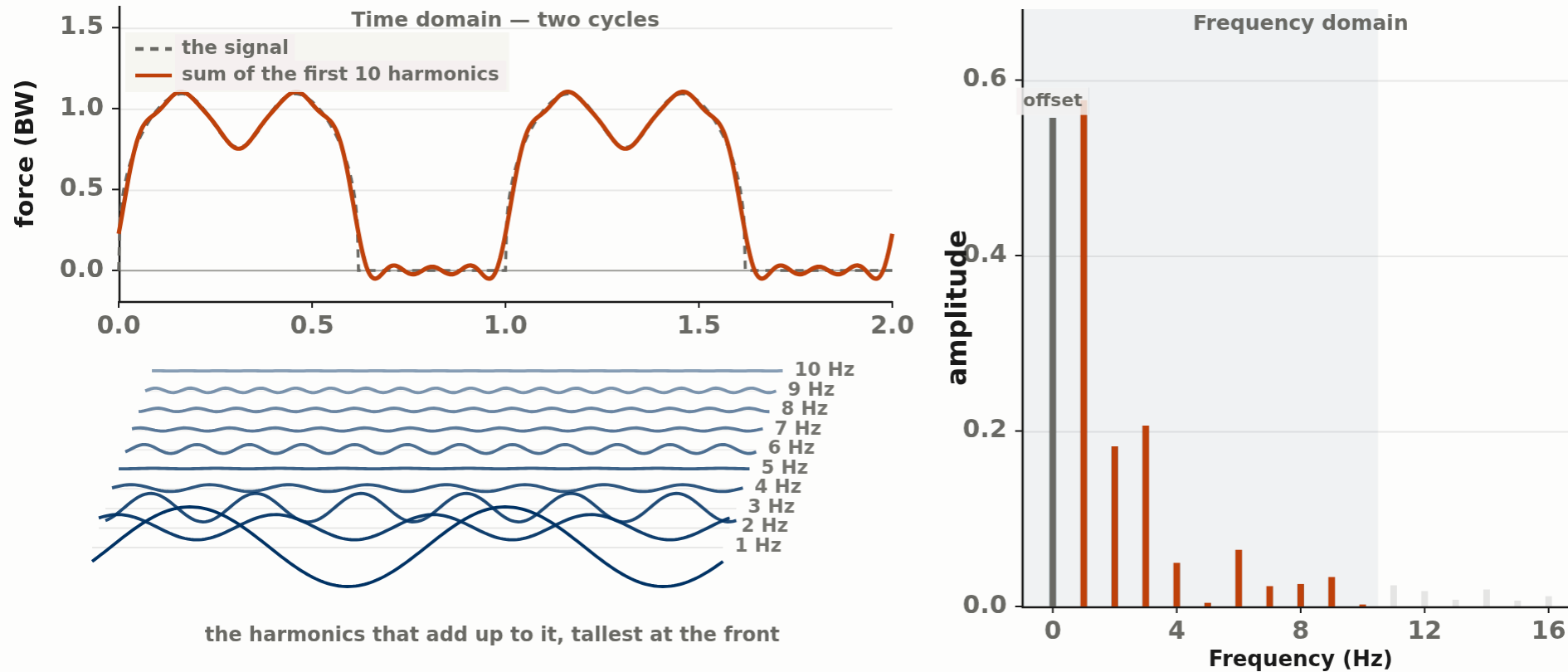
1 Hz

5 Hz

2 Hz + 30 Hz

25 + 50 Hz

Fourier transform



$$x(t) \approx 0.56 - 0.21 \cos(2\pi \cdot 1t) + 0.54 \sin(2\pi \cdot 1t) + 0.13 \cos(2\pi \cdot 2t) + 0.13 \sin(2\pi \cdot 2t) - 0.19 \cos(2\pi \cdot 3t) + 0.09 \sin(2\pi \cdot 3t) \dots$$

10 harmonics — every whole multiple of the 1 Hz fundamental up to **10 Hz**. Residual RMS **0.037**. Six to eight harmonics carry almost all of a walking GRF — which is where the usual 6 Hz cutoff comes from.

Walking GRF

Running GRF

Square wave

one wave per harmonic

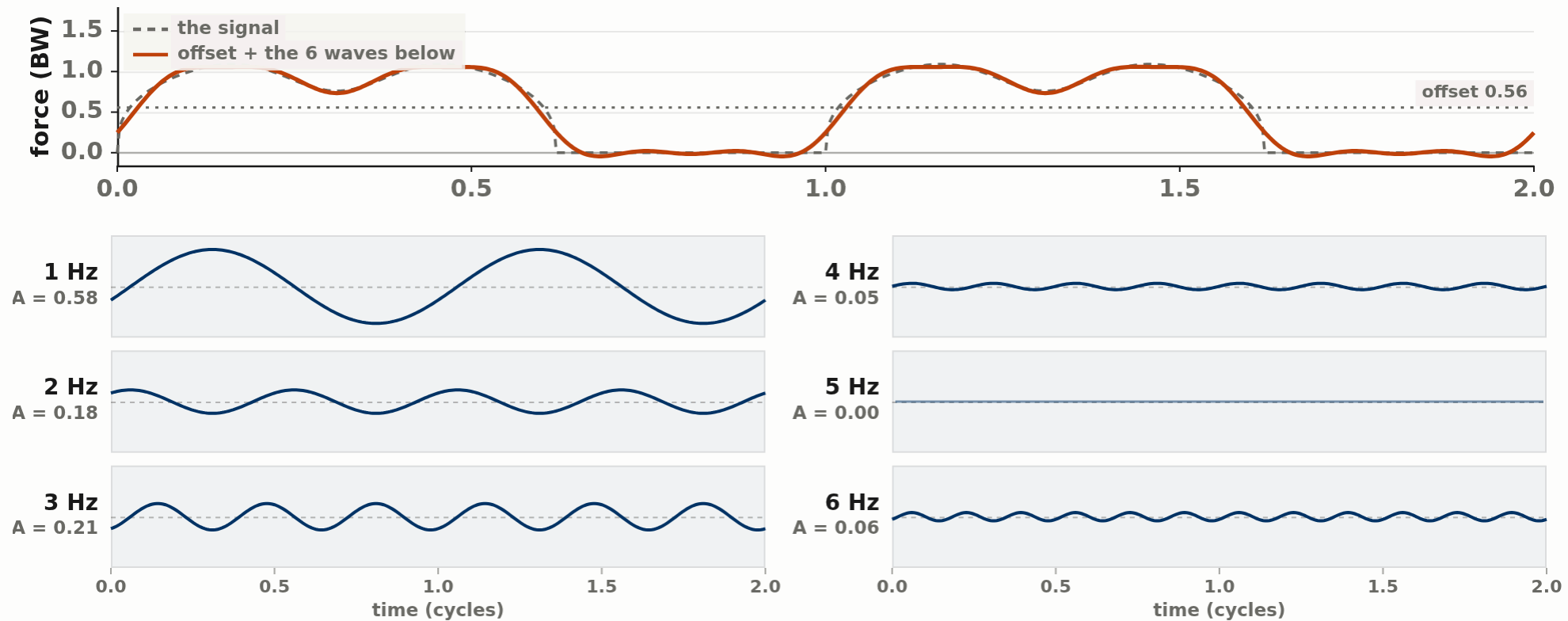
sin & cos parts

► Build up

Harmonics kept

10 of 16

The components, one graph each



Walking GRF = 0.56 offset + the 6 waves below, added together

One flat graph per harmonic. One shared amplitude scale, so a term drawn flat really is contributing almost nothing.

Walking GRF

Running GRF

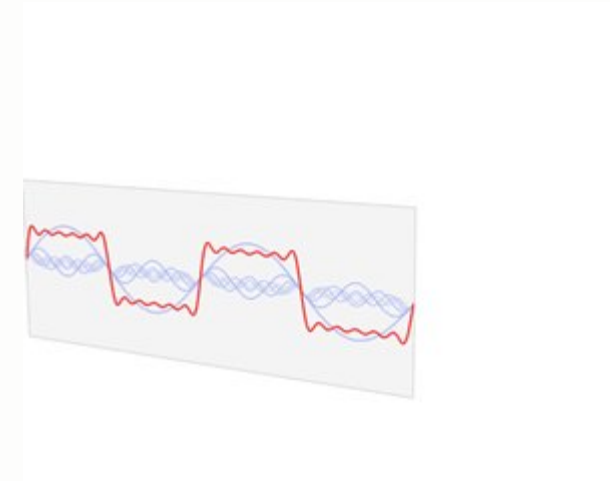
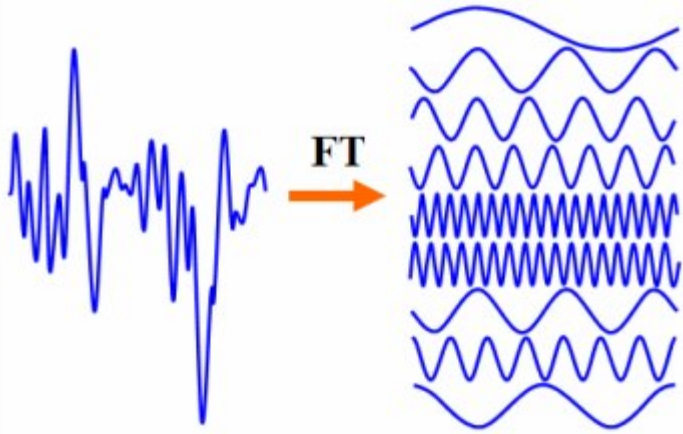
Square wave

one wave each

sin & cos

one scale

each to fit

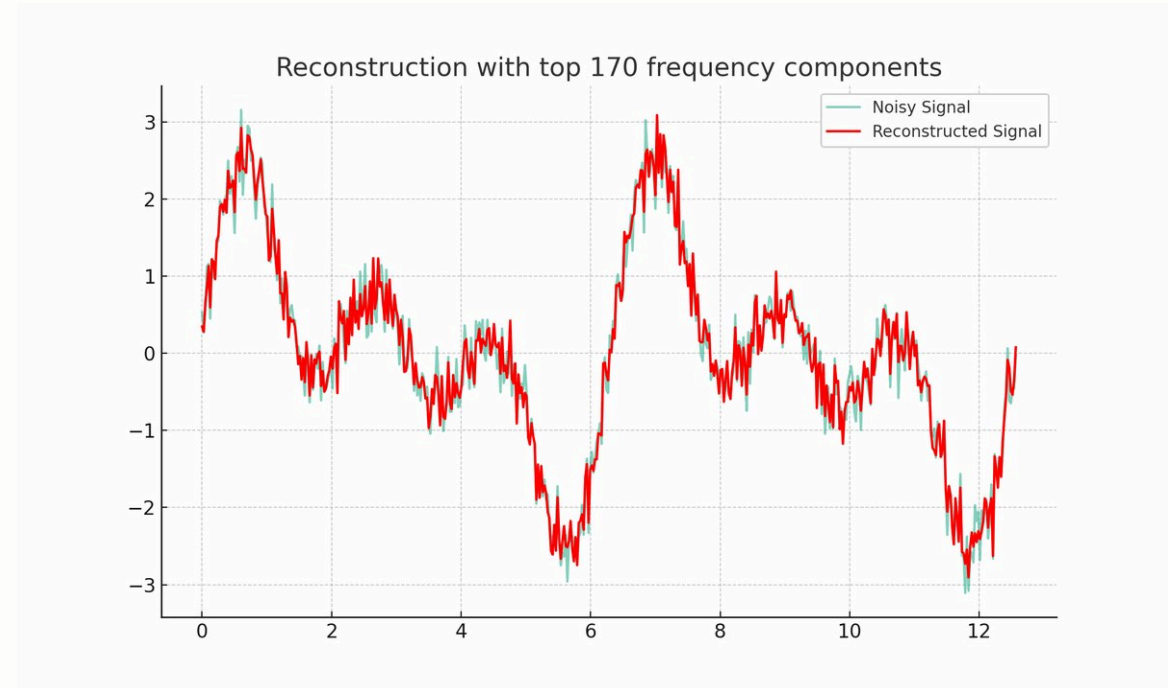


Digital filtering

Is a way of reducing the frequency content of the signal so that it just obtains the signal you are interested in without the noise.

A common use of Fourier transforms is to find the frequency components of a time-domain signal that is mixed with noise.

Because noise is often of a different frequency than the intended signal, a filter can be used to remove the noise (ie: parts of the signal that are a certain frequency).



Types:

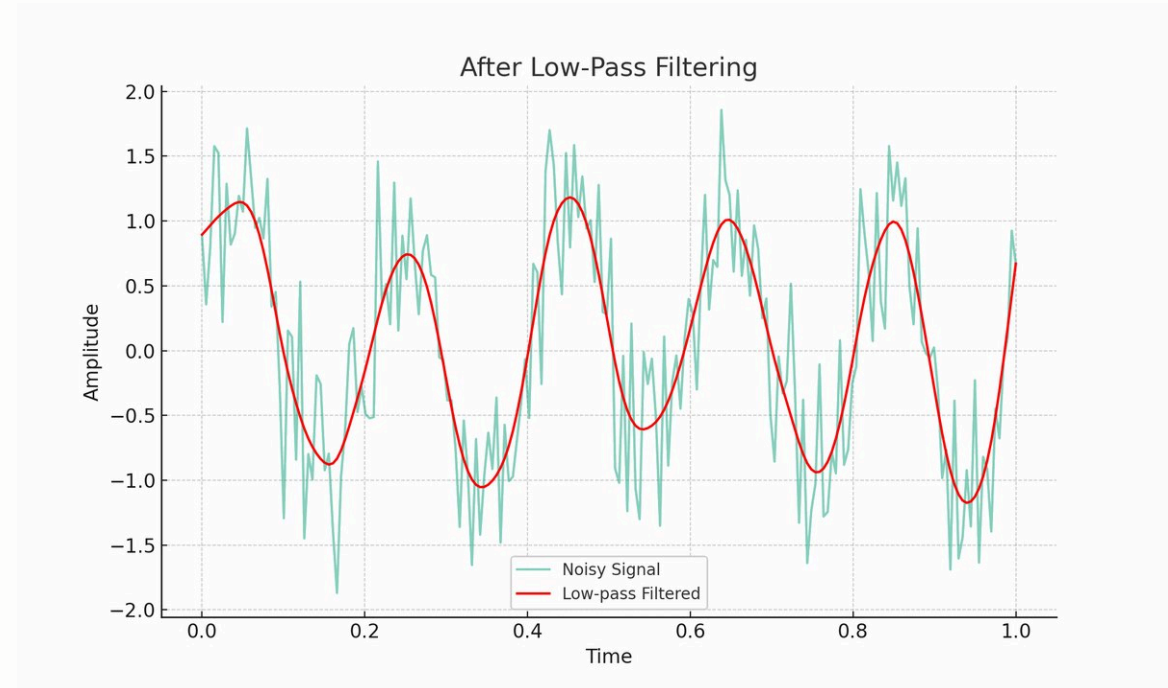
low-pass, high-pass, band-pass, band stop and notch (single or small band-stop filter, useful for AC interference)

Designs:

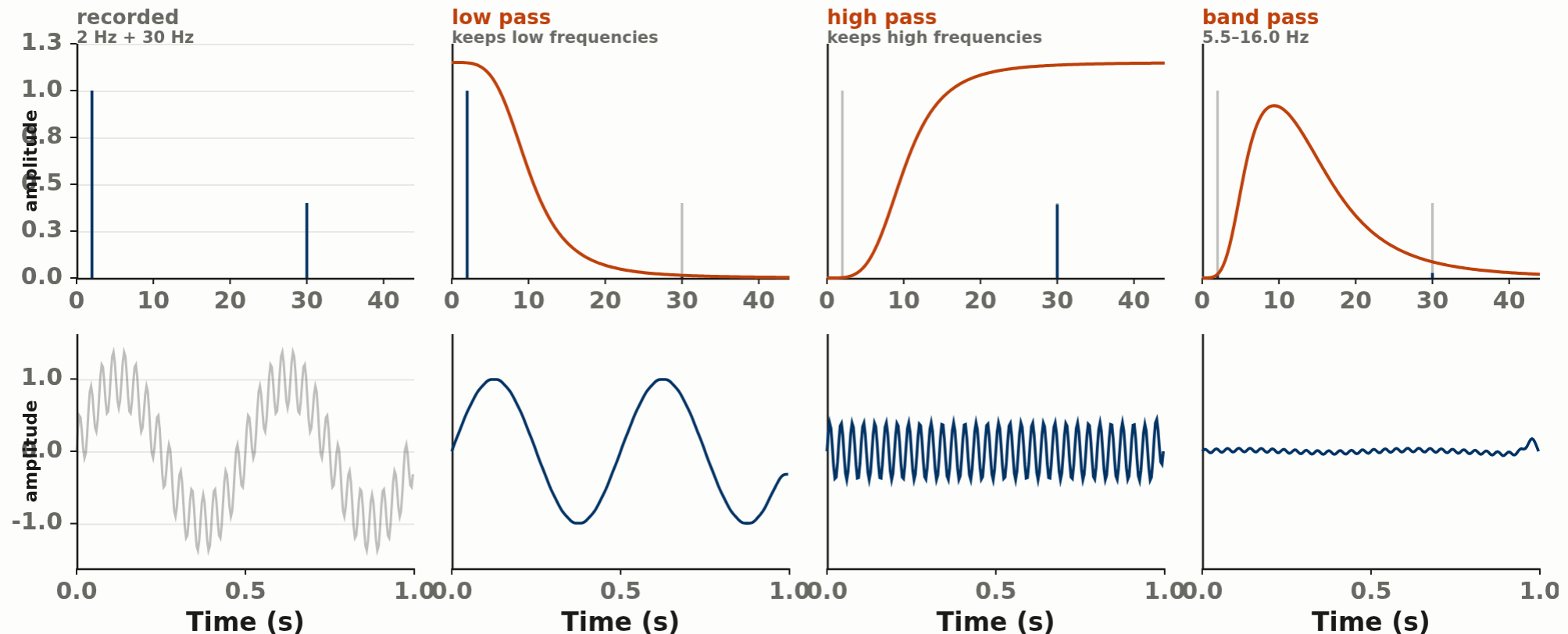
Butterworth, critically-damped, Chebyshev etc. (sharper cutoffs), Generalized Cross-validation (GCV also called Woltring filter, 1986)

Problems:

Phase distortion (usually phase-lags occur)



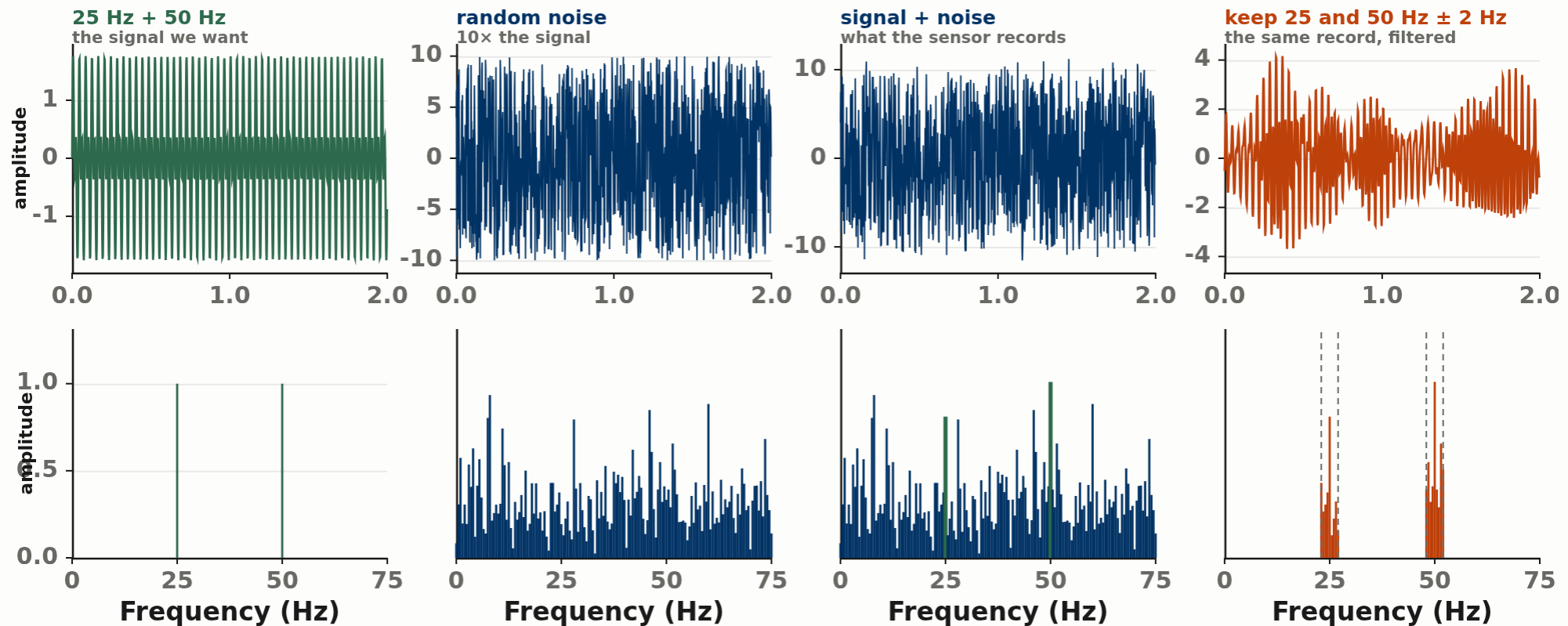
Let's say that we wanted to isolate different parts of the signal — we could use different filters to do that. A **low pass filter** maintains low frequencies; a **high pass filter** maintains high frequencies.



cut-off **10 Hz** · low pass keeps **93%** of the amplitude, high pass keeps **37%**

The **red line** is the filter's own response — what fraction of each frequency it lets through. Grey spikes are what was recorded, blue what survived. Slide the cut-off past 2 Hz and the low-pass column throws away the signal it was meant to keep, which is the whole argument for choosing a cut-off from the signal rather than by habit.

25 Hz + 50 Hz, buried in random noise **ten times larger** than the sine waves. In the time domain there is nothing left to see. In the **frequency domain** the two spikes are still standing there — which is what lets a filter pull the signal back out.



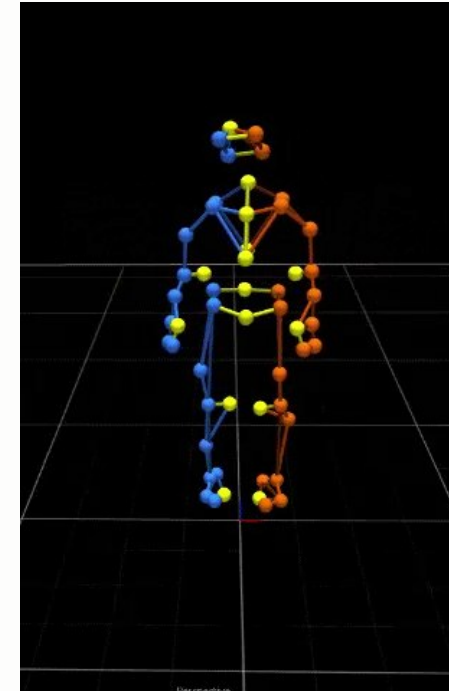
noise **10×** the signal · in the buried record the 25 Hz spike still stands at **0.81** against an average noise floor of **0.33** · after filtering, the record matches the clean signal to **r = 0.62**

Digital filtering

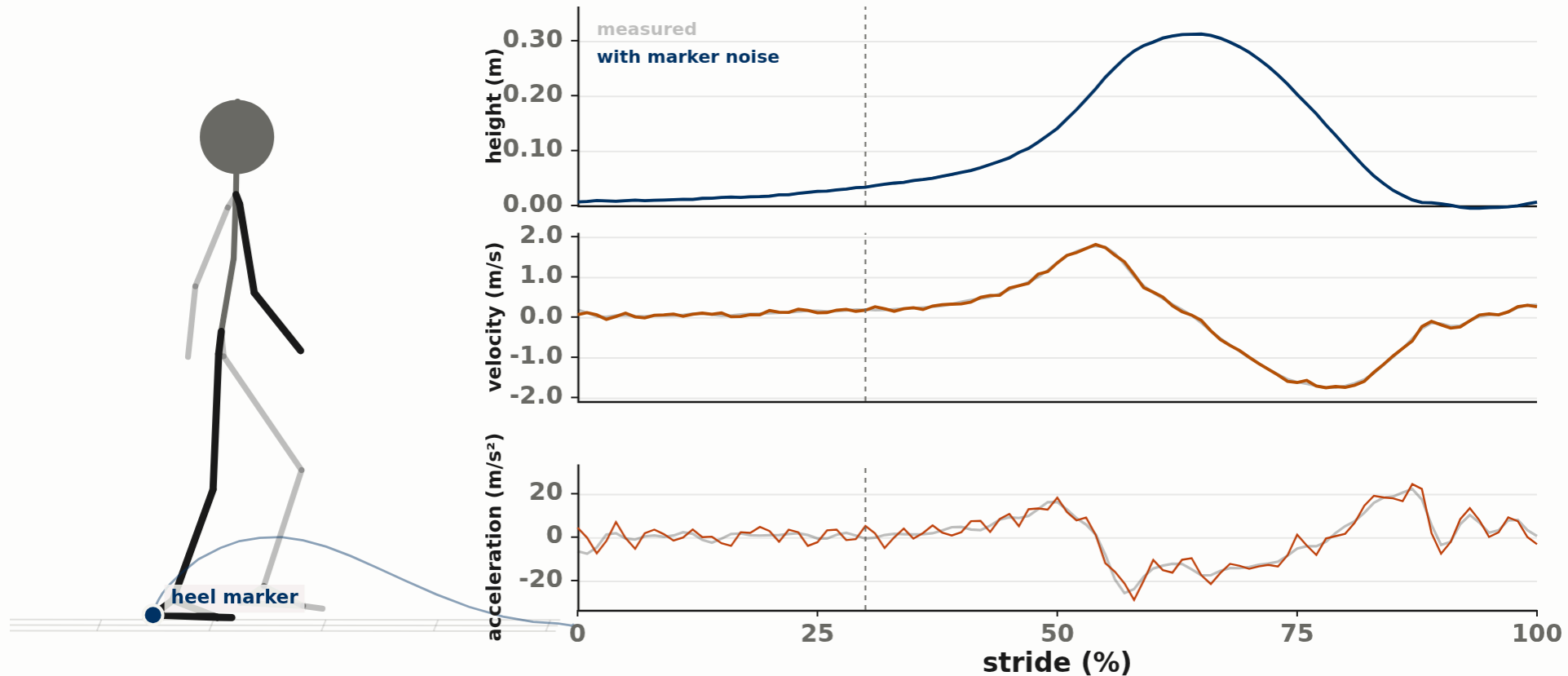
The most common source of noise in human motion capture data are small errors in the position of the biomechanical markers during the digitization process.

This noise is typically **low amplitude and high frequency**; therefore, a low-pass filter is often used to smooth kinematic data.

The aim of filtering is to arrive at the highest **signal : noise** ratio



Noise in motion capture data is **amplified** each time the data are differentiated or integrated.



right heel, one stride, sampled at **90 Hz** · marker noise ± 1.0 mm · peak acceleration **29 m/s²** against a true **26 m/s²**

A millimetre is nothing in the height trace — it is thinner than the line. But a difference divides by dt , and doing it twice divides by dt^2 , so the same millimetre comes out as metres per second squared. Step the camera up to 360 Hz and the movement does not change at all, while the noise in the acceleration gets **sixteen times worse** — the faster camera samples the same marker error more often and across a shorter interval. Press the low-pass and the acceleration comes back, which is why kinematic data is filtered **before** it is differentiated, not after — and why the cut-off matters: drop it to 6 Hz and the filter starts shaving the real peak off too.

Digital filtering

Filtering, smoothing and fitting can aid in signal **Event Detection**.

In many biomechanical analysis, signal processing algorithms are required.

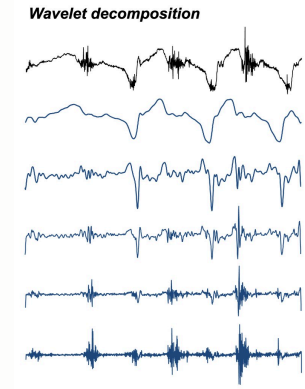
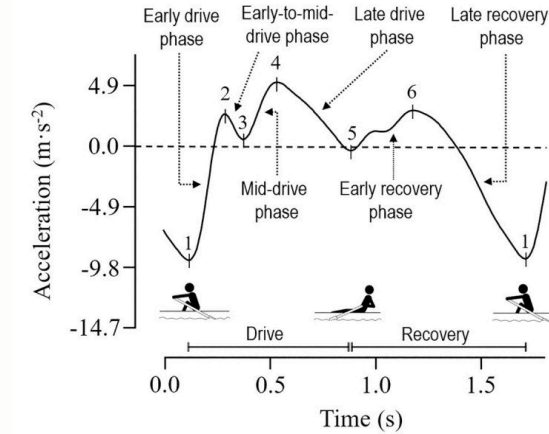
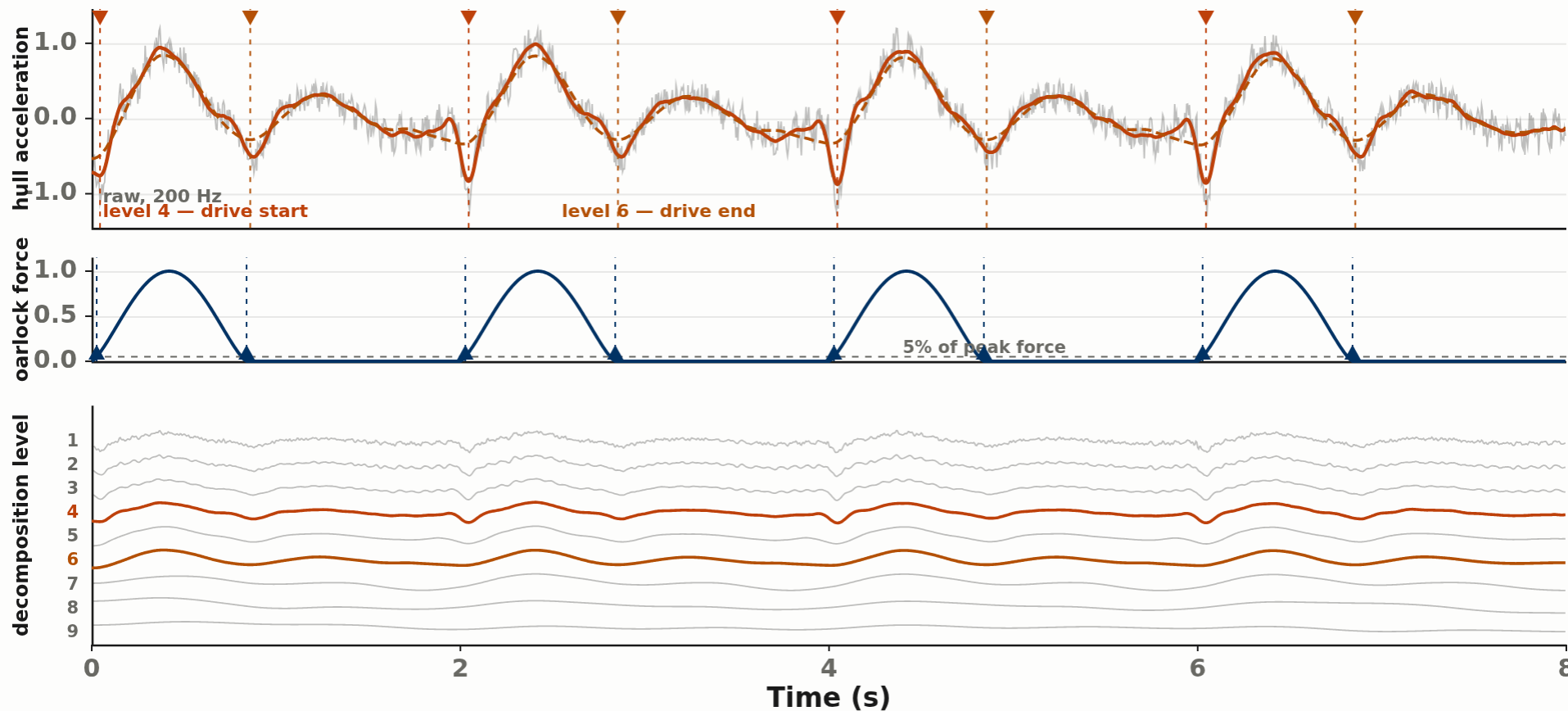


Figure 2. 5 level wavelet decomposition of hull acceleration signal.

Undecimated wavelets on a hull accelerometer — finding the drive in the noise



drive start from level 4 is out by 18 ms · drive end from level 6 is out by 16 ms, against the oarlock force

The raw trace has the landmarks in it, but it also has hull ring and sensor noise, so a peak detector run on it finds dozens of peaks per stroke. Each level of the wavelet transform is a smoother version of the same signal, and because the transform is **undecimated** every level stays the same length — an event found at level 6 still sits at the sample it happened at. Too low a level and the noise is still there; too high and the landmark slides away from where it happened. Geneau et al. (2024) settled on different levels for the two events and reported ICCs of 0.88–0.94 for drive time against an instrumented oarlock.

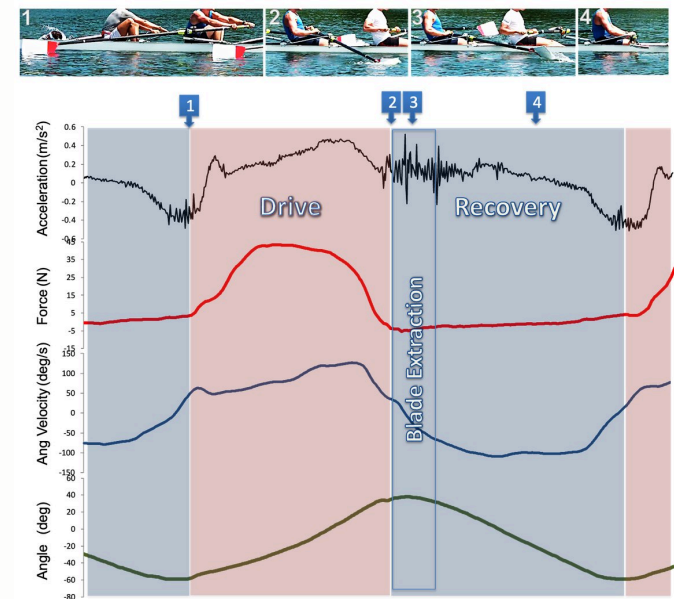
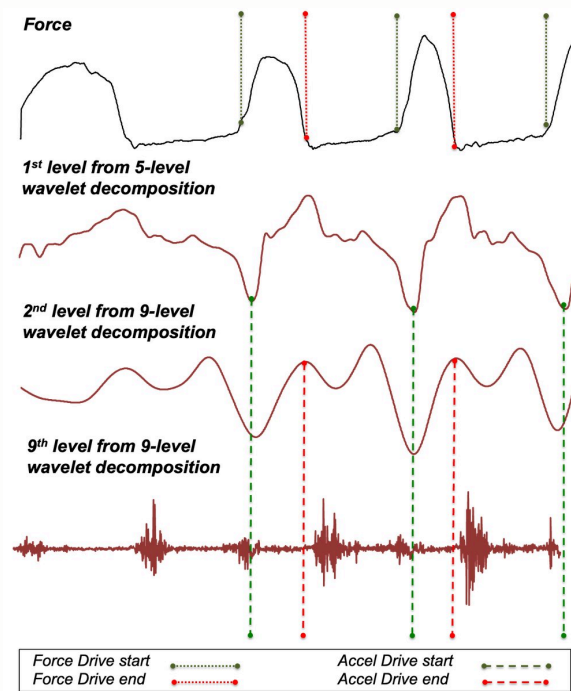


Figure 1. Rowing hull acceleration, force, oar angle and angular velocity data showing important phase features.

- It is important to know features of the signal we are collecting.
- Highest natural frequency in signal helps define the **minimum sampling rate**.
- Data processing can come in the form of **Filtering, smoothing and fitting**.
- It is important to get rid of noise **before and after** data transformation.